

Video Deflector Driver Test Procedure

CLS TECHNICAL PROCEDURE – 1.7.27.1 Rev 2

Date: 2011-06-08

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Testing the Video Deflector Driver

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REVISION HISTORY

<i>Revision</i>	<i>Date</i>	<i>Description</i>	<i>Author</i>
A	2001-02-23	Original Draft	N.G. Johnson
0	2001-04-25	Released	N.G. Johnson
0A	2009-10-28	Added Scope Picture	D. Beauregard
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1. Purpose

This document describes the procedure to verify the operation of a new driver for the video deflector plates or to diagnose a failing driver.

2. Background

See CLS document 1.2.46.1 Video Deflector Modification.

3. Definitions and Abbreviations

HV High Voltage

4. References

CLS documents:

/ACCL/EE/VD/0046100 Video Deflector Driver

/ACCL/EE/VD/0046110 Video Deflector Wiring

/CDAC/EE/TMNG/FORP/004170 Fibre Optic Receiver for Video Deflector

5. Required Equipment

- Digital storage oscilloscope with 50 Ω inputs rated for >250 MHz bandwidth
- Adjustable high voltage power supply 0-1800 V, current limited to <10 mA.
- High voltage power supply output cable with SHV connector, positive voltage on center pin
- Grounding hook with insulated handle for discharging residual high voltage
- Multimeter
- 5.2 VDC power supply
- 5.2 V power cable with Amphenol MS3106A connector.
- Fibre optic pulse generator
- Fibre optic cable
- HV step down divider (ACCL/EE/VD/0046100/HV Step Down Box/0048101) OR HV Oscilloscope Probe

6. Procedure Steps

Refer to Technical Procedure 7.7.39.5 Testing the Components of the Fibre Optic Receiver System if the fibre optic receiver has not been tested.

6.1 High Voltage Driver

Note: When using the high voltage power supply insure the output voltage is set at or below the voltage required before turning it on for one of the steps listed below. If in doubt the power supply should be set to 0 Volts before turning it on. Always ground high potential points after switching the high voltage power supply off. Normally grounding the hot side of C1 will be sufficient, but a board with defective components could still have high voltage at the high voltage input or at the high voltage pulse output. If connections are made, or disconnected the high voltage power supply should also be unplugged. These tests should only be performed by individuals familiar with procedures associated with high voltage circuits operating up to 2 kV.

The high voltage supply must be switched off, high potential points must be grounded, and the high voltage supply must be unplugged before the circuit is left unattended.

Be alert for indications of malfunction such as arcing or burning.

If a malfunction or test failure occurs stop the test procedure, turn off the high voltage, ground the high potential points, and record the failure.

Do not use oscilloscope probes to monitor high voltage points. Arcing as the probe approaches or is withdrawn will damage the circuit and requires transistor replacement.

All voltage measurements are with respect to the ground on the avalanche transistor board.

6.1.1 Power

1. If testing a new board, inspect the circuit visually for correct component placement, solder bridges and missing solder joints before applying power.
2. Connect the monitor output to the oscilloscope input. Connect the HV power supply to the high voltage input connector. Set the voltage to 200 V. The monitor output has a 1000:1 divider. Verify that the monitor output is 200 mV, ± 10 mV.
3. Gradually increase the high voltage to 1800 V. The monitor output voltage should be $1.7 \text{ V} \pm 150 \text{ mV}$. Record this voltage.
4. Switch off the high voltage supply, ground the hot side of C1, and unplug the supply.
5. Connect the 5 V DC supply to the fibre optic receiver board. Connect the fibre optic cable from the fibre optic pulse generator. Adjust the pulse generator to produce a $1 \mu\text{s}$ pulse at a rate of 10 Hz.
6. Verify that a 0.7 – 0.8 Volt positive voltage pulse is observed at the 'In+' input.
7. Turn on the high voltage power supply, which still is set to 1800 V.
8. Verify that a regular "ticking", similar to the sound of a 0.5 mm pencil lead breaking, can be heard at the trigger frequency.
9. Observe the monitor output and verify that the voltage drops greater than 0.1 V as the circuit fires, then returns to the level noted in step 3. See figure1.
10. Adjust the pulse width from the pulse generator and verify that the trigger pulse is seen not to move with the trailing edge.
11. Switch off the high voltage supply, ground the hot side of C1, and unplug the supply.

12. Connect the high voltage step down divider and view the waveform with an oscilloscope and ensure that the pulse shape appears similar to the one shown in Figure 1 (below).

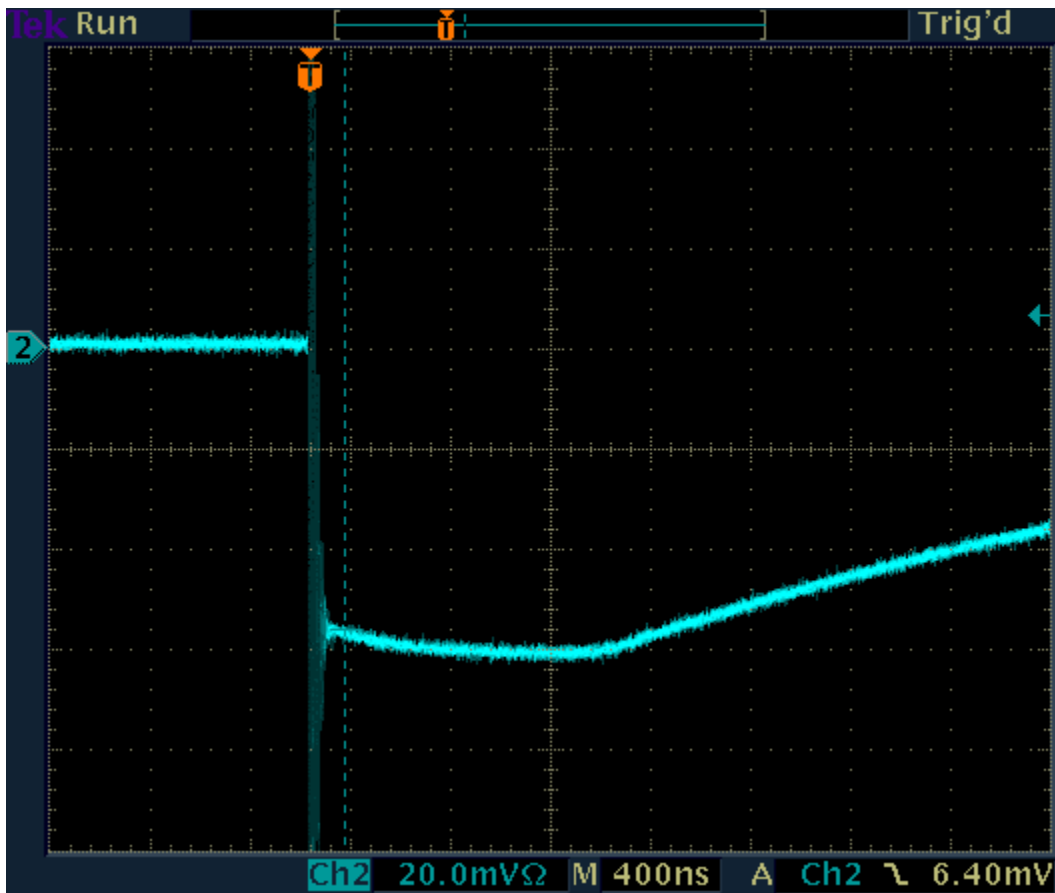


FIGURE 1

Appendix A- Test Sheet for Video Deflector Driver - ACCL/EE/VD/0046100

Check box for every test completed. If test is not applicable, write N/A. If testing more than 10 boards, use additional test sheet.

Board	1	2	3	4	5	6	7	8	9	10
Serial No.										
6.1.1 Power										
1.										
2.										
3.										
Voltage										
4.										
5.										
6.										
7.										
8.										
9.										
10.										
11.										
12.										

Date: _____ Tested by: _____ Signature: _____
(YYYY-MM-DD) (print name)